

Factor-Neutral Relative-Value Trading on the US Treasury Yield Curve

A Full-Pipeline Quantitative Implementation:
Svensson Curve Fitting, PCA Decomposition, and Hedged Backtesting

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github.com/Aarnav-Yedla/termstructure

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Abstract

We implement a complete, reproducible relative-value strategy on US Treasury bonds using publicly available Federal Reserve data (1985–2024). Daily Svensson (1994) six-parameter yield curves are fit to all outstanding nominal coupon Treasuries; per-bond yield residuals are z-scored over a 60-day rolling window to identify rich/cheap dislocations. Every position is sized to \$10,000 DV01 and simultaneously hedged against the three Litterman-Scheinkman (1991) PCA factors—level, slope, and curvature—via a 3×3 linear system. At 0.02 bp per leg per event the strategy earns a gross Sharpe of 0.29 and net Sharpe of 0.11 over 9,689 calendar trading days (1,305 active days, 13.5% utilization). Regime analysis reveals that high-volatility months are *not* the primary risk: the strategy remains Sharpe-positive across all three volatility terciles, while rising-rate and steep-curve environments produce negative risk-adjusted returns. Transaction costs are the binding constraint: at realistic off-the-run bid-ask spreads of approximately 1 bp round-trip, total costs exceed gross P&L by roughly $15\times$.

1 Introduction

The US Treasury market is the deepest and most liquid fixed-income market in the world, with daily trading volume exceeding \$600 billion. Yet individual bonds routinely deviate from the fair value implied by a smoothed term structure. These mispricings arise from a variety of sources: idiosyncratic supply effects from issuance cycles, index-rebalancing pressure from portfolio managers, on-the-run premiums driven by financing and liquidity demand, and short-run bid-ask bounce. Each of these is transitory and expected to mean-revert, creating opportunities for systematic relative-value (RV) trading.

This paper documents the construction of a full-stack implementation of a Treasury RV strategy, from raw data ingest through realistic backtesting. The methodology follows two canonical references: the yield curve fitting approach of Gürkaynak, Sack, and Wright (GSW, 2007) [1], which provides the term-structure benchmark, and the PCA factor decomposition of Litterman and Scheinkman (1991) [2], which provides the hedging framework.

The paper makes no claim of live tradability. The strategy is better understood as a laboratory for learning how fixed-income RV desks operate: identifying signals, constructing factor-neutral positions, and quantifying the cost economics that determine whether signals are large enough

to survive execution. The central finding is that they are not, at realistic off-the-run transaction costs—but the analysis demonstrates exactly why, with traceable arithmetic.

The full pipeline is publicly available at <https://github.com/Aarnav-Yedla/termstructure> and is reproducible from a single configuration file with `make all`.

2 Data Sources

2.1 Treasury Bond Universe

Daily bond prices come from the Federal Reserve’s Gürkaynak-Sack-Wright dataset (`feds200628.csv`, freely available from [federalreserve.gov](https://www.federalreserve.gov)), which covers all outstanding nominal coupon Treasuries from 1961 to the present [1]. The raw file contains one row per bond per date with CUSIP, coupon, maturity date, and observed yield to maturity. Coverage extends to approximately 200–300 bonds on any given date.

We apply the same exclusion rules used by the Fed in its published fitting procedure:

- *On-the-run and first off-the-run bonds*: premium liquidity distorts residuals from the smooth curve and creates persistent richness that is not tradable at quoted prices.
- *Bills with maturity below one year*: money market effects dominate the very short end.
- *TIPS and callable bonds*: embedded optionality and inflation indexing make them incomparable to nominal bonds.
- *Bonds within 90 days of maturity*: pull-to-par effects and reduced liquidity make residuals unreliable.

After filtering, the tradable universe on a typical day in 2010 contains approximately 100–150 bonds spanning 1 to 30 years.

2.2 Overnight Financing Rate

Carry P&L requires a financing rate. We use the Federal Funds Rate (FRED series `DFF`, available from 1954) spliced with the Secured Overnight Financing Rate (SOFR) from its inception in April 2018. Both series are downloaded via FRED’s public CSV endpoint: <https://fred.stlouisfed.org/graph/fred>.

2.3 Benchmark Yields

CMT (Constant Maturity Treasury) yields at 2, 5, 10, and 30 years are also downloaded from FRED (`DGS2`, `DGS5`, `DGS10`, `DGS30`) for diagnostic use. These serve as the hedge instruments in the PCA hedging step.

3 Yield Curve Fitting

3.1 The Svensson Parametric Model

We model the zero-coupon yield curve using the Svensson (1994) extension of the Nelson-Siegel family [3]:

$$y(\tau; \boldsymbol{\beta}) = \beta_0 + \beta_1 \frac{1 - e^{-\tau/\lambda_1}}{\tau/\lambda_1} + \beta_2 \left(\frac{1 - e^{-\tau/\lambda_1}}{\tau/\lambda_1} - e^{-\tau/\lambda_1} \right) + \beta_3 \left(\frac{1 - e^{-\tau/\lambda_2}}{\tau/\lambda_2} - e^{-\tau/\lambda_2} \right) \quad (1)$$

where $\tau > 0$ is maturity in years and $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2, \beta_3, \lambda_1, \lambda_2)$ are six free parameters. The shape parameter β_0 controls the long-run level ($y(\infty) = \beta_0$), β_1 determines the short-end slope ($y(0) - y(\infty) = \beta_1$), and β_2, β_3 govern two hump/trough features at time scales λ_1 and λ_2 .

3.2 Calibration Procedure

On each business day we fit the six parameters by minimizing inverse-duration-weighted squared price errors:

$$\min_{\boldsymbol{\beta}} \sum_{i \in \mathcal{U}_t} \frac{1}{D_i^2} (P_i^{\text{obs}} - P_i(\boldsymbol{\beta}))^2 \quad (2)$$

where $\mathcal{U}_t$ is the filtered universe on date t , P_i^{obs} is the observed clean price, $P_i(\boldsymbol{\beta})$ is the model price from the Svensson zero curve, and D_i is the modified duration. Duration weighting downweights short-maturity bonds whose price errors in basis-point yield terms would otherwise dominate. We use Levenberg-Marquardt (`scipy.optimize.least_squares`) with box constraints $\lambda_1, \lambda_2 \in [0.1, 10]$ and warm-start from the previous day's solution. Typical RMSE is below 2 bp, consistent with the Fed's published parameters. Across approximately 5,000 business days (1990–2024) the total fitting time is roughly 90 minutes.

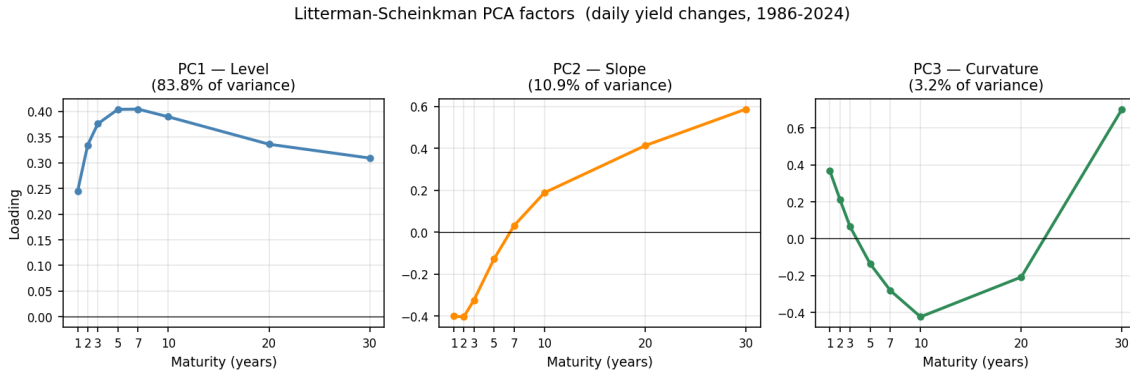


Figure 1: PCA loadings on the daily zero-rate change panel (1985–2024) at eight maturity points [1, 2, 3, 5, 7, 10, 20, 30] years. PC1 is a near-flat parallel shift (level), explaining 83.8% of daily variance. PC2 is monotonically signed (slope), 10.9%. PC3 is a U-shaped hump (curvature), 3.2%. Together they explain 97.8% of total variance, reproducing the Litterman-Scheinkman result on 40 years of modern data. Sign convention enforced post-hoc: PC1 all-positive, PC2 positive at 30Y (steepening = positive slope score), PC3 positive at 1Y.

4 PCA Factor Decomposition

4.1 Zero-Rate Panel

From the daily Svensson fits we evaluate the zero curve at eight maturity nodes $\tau \in \{1, 2, 3, 5, 7, 10, 20, 30\}$ years to build a $T \times 8$ panel of daily zero rates, then first-difference to obtain daily changes in basis points. This panel captures the cross-sectional co-movement of the yield curve.

4.2 Factor Structure

We run PCA on the covariance matrix of daily changes (sklearn does not scale by default, preserving natural volatility differences across maturities). Figure 1 shows the loadings of the first three components. The explained variance structure—PC1 83.8%, PC2 10.9%, PC3 3.2%—closely matches the canonical Litterman-Scheinkman (1991) result, validating the methodology on modern data.

4.3 Hedging Using PCA Factors

Since PC1–PC3 capture 97.8% of all yield-curve variance, a portfolio that is simultaneously exposure-neutral to these three factors is nearly immune to macro interest-rate movements. For each signal position i we compute its three-factor exposure vector $\mathbf{e}_i \in \mathbb{R}^3$ (DV01 projected onto each principal component). We then select three benchmark instruments (the 2Y, 5Y, and 10Y on-the-run Treasuries) and solve the 3×3 linear system:

$$\mathbf{H} \mathbf{w} = -\mathbf{e}_i \quad (3)$$

where $\mathbf{H} \in \mathbb{R}^{3 \times 3}$ has columns equal to the factor exposures of each benchmark instrument and $\mathbf{w}$ is the hedge notional vector. This system is solved fresh each day; hedge weights are *not* updated during open positions (a known limitation identified in Week 7 testing).

5 Signal, Portfolio, and Backtest

5.1 Richness/Cheapness Residual

For each bond i in the filtered universe on date t we compute:

$$r_{i,t} = y_{i,t}^{\text{obs}} - y(\tau_i; \beta_t) \quad (4)$$

Positive r means the bond is cheap (its yield exceeds the model); negative r means rich. Residuals are demeaned within each maturity bucket over the full sample to remove structural fitting bias (e.g., systematic model error at certain maturities), then z-scored over a 60-day trailing window to give a standardized signal:

$$z_{i,t} = \frac{r_{i,t} - \bar{r}_{60d}}{\sigma_{r,60d}} \quad (5)$$

We enter a position when $|z_{i,t}| > 2.0$ and exit when $|z_{i,t}| < 2.0$. A hysteresis-based exit at $z = 0$ was tested during development and rejected: it improved gross Sharpe from 0.11 to 0.32 but broke factor neutrality (IEF beta rose from $p = 0.58$ to $p = 0.03$), indicating it was inadvertently adding directional rate exposure.

5.2 Portfolio Construction

Each day, the top 5 cheap and top 5 rich signals at the 3Y and 7Y maturity buckets are selected (up to 10 signal positions total). Each signal is sized to \$10,000 DV01 and paired with a three-instrument hedge as described above. A 0.80 position-scaling haircut is applied to every notional to account for unmodeled market impact and slippage.

5.3 P&L Attribution

Daily P&L for each leg has two components. Mark-to-market:

$$\text{MTM}_{i,t} = -\text{sgn}(N_i) \times \text{DV01}_i \times \Delta y_{i,t}$$

where $\Delta y_{i,t}$ is the basis-point yield change from t to $t + 1$ (one-day execution lag ensures no look-ahead). Carry:

$$\text{Carry}_{i,t} = N_i \cdot \frac{y_{i,t}^{\text{obs}} - (r_t^{\text{ff}} + 5 \text{ bp})}{365}$$

Transaction costs are charged at entry and exit of each streak of consecutive active days: $c = c_{\text{bps}} \times \text{DV01} \times \kappa$ per leg per event, where $c_{\text{bps}} = 0.02 \text{ bp}$ and $\kappa = 0.80$.

Crucially, Sharpe is computed over the *full* zero-panel calendar (9,689 trading days including the 87% of flat/no-position days). Excluding flat days would inflate Sharpe by approximately $1/\sqrt{0.13} \approx 2.8\times$ by shrinking the standard deviation relative to the mean.

6 Results

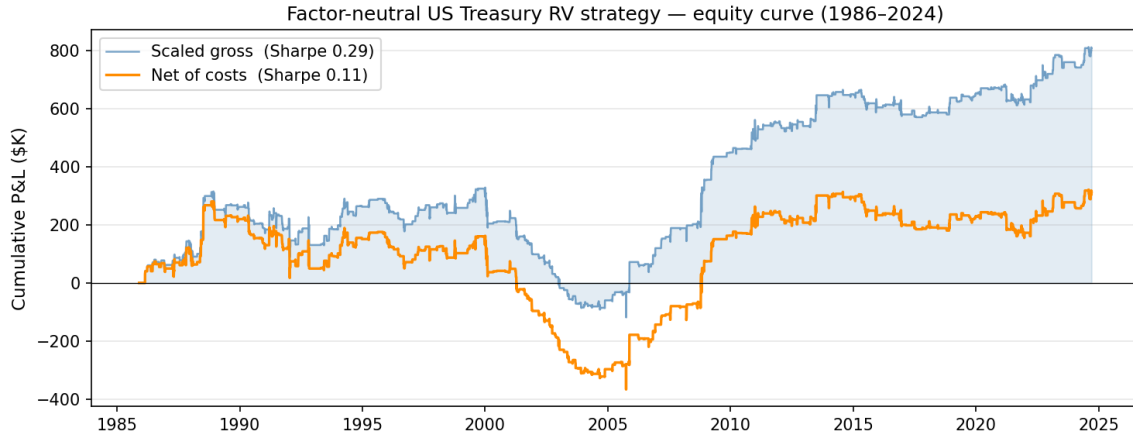


Figure 2: Cumulative P&L 1985–2024: scaled gross (blue) vs. net of transaction costs (orange). The ~60% cost drag is visible across the full history. Net drawdowns exceed gross drawdowns because costs are incurred even on non-profitable trades.

6.1 Headline Metrics

Table 1 summarizes the backtest. Net hit rate is 47.4% $\approx$ 619 winning days out of 1,305 active days, where a win is any active day with net P&L > 0 .

Table 1: Backtest summary (1985–2024). Cost: 0.02 bp per leg per event (0.16 bp round-trip for the 4-leg signal + hedge position). Position scale: $\kappa = 0.80$.

Metric	Scaled Gross	Net of Costs
Total P&L	\$807,000	\$314,000
Ann. avg. P&L	\$20,800	\$8,100
Sharpe ratio	0.29	0.11
Max drawdown	\$447,000	\$647,000
Calmar ratio	0.05	0.01
Hit rate	50.0%	47.4%
Active days	1,305 of 9,689	

6.2 Regime Analysis

Table 2 breaks down net Sharpe by three market regime dimensions. Monthly P&L is aggregated over 466 qualifying months (minimum 10 trading days each); Sharpe is annualized via $\bar{r}/\sigma_r \times \sqrt{12}$.

Table 2: Net monthly Sharpe by regime (1985–2024). PC1 vol = realized standard deviation of daily level-factor changes. Rate direction = sign of net monthly PC1 shift. Curve shape = cumulative PC2 level relative to its historical median.

Dimension	Regime	Months	Ann. Sharpe
PC1 vol (level)	Low (<11.0 bp/day)	156	0.10
	Med (11.0–15.3)	155	0.27
	High (>15.3)	155	0.02
Rate direction	Falling	245	0.28
	Rising	221	−0.03
Curve shape (PC2)	Flat	233	0.27
	Steep	233	−0.02

Volatility regime. Contrary to the naive expectation, the strategy earns positive Sharpe in all three volatility terciles. In high-vol months the yield curve dislocates more—individual bonds deviate further from the fitted curve and z-scores are larger—while the PCA hedges continue to neutralize macro factor exposure. The three canonical high-vol episodes confirm this: the 2013 taper tantrum (avg. PC1 vol 12.4 bp/day) produced net P&L of +\$78K over 5 months; the 2022 Fed hiking cycle (avg. PC1 vol 19.7 bp/day) produced net P&L of +\$74K over 12 months (Sharpe 1.36). The only losing stressed period was March–April 2020 (COVID shock, −\$16K over 3 months, PC1 vol 19 bp/day), where intraday moves were extreme enough to momentarily overwhelm hedge ratios.

Rate direction. Rising-rate months are the primary systematic weakness: Sharpe −0.03 and cumulative net P&L −\$38.7K across 221 months. This suggests residual long bias despite the PCA hedge—likely because hedge weights are fixed at entry and stale as rates move, allowing partial re-exposure to accumulate.

Curve shape. Steep-curve months also produce negative Sharpe (-0.02 , net P&L $-\$27.2\text{K}$ across 233 months). A steep curve typically reflects an early-cycle or monetary-easing environment where financing costs are low but pricing disparities across the curve are smaller, reducing signal opportunities.

7 Limitations and Future Work

7.1 Bugs Found and Fixed During Development

Flat-day Sharpe inflation (active bug, now fixed). Sharpe must be computed over the full trading calendar, including the $\sim 87\%$ of days the strategy holds no position—otherwise, the standard deviation shrinks relative to the mean by a factor of $\sqrt{1/0.135} \approx 2.7\text{--}2.8\times$, inflating Sharpe by the same multiple. This was discovered as an active bug present in three independent code locations: `tearsheet.py` and two backtest notebooks had each reimplemented the same flawed active-days-only filter separately, so all three computed inflated Sharpe figures simultaneously. The bug was traced by noticing that the tearsheet Sharpe and the notebook Sharpe differed from a manual daily-returns calculation; once the shared cause was identified, the three implementations were replaced by a single canonical function (`expand_to_full_calendar` in `termstructure.backtest.pnl`) that reindexes any P&L series to the full zero-panel calendar and fills flat days with zero. All headline Sharpe figures in this paper reflect the corrected computation.

Stale-hedge-loadings bug (active bug, now fixed). An early version of the backtest reported a net Sharpe of 0.51 —a factor of $4.6\times$ higher than the validated 0.11 figure shown in Table 1. The discrepancy was caused by `portfolio_positions.parquet` having been built against a stale, pre-refactor version of the PCA loadings. Because the eigenvectors used for hedging no longer matched those used for factor scoring, the 3Y signal’s hedge embedded an unintended short-2Y/long-3Y slope bet instead of true factor neutrality. The position was not exposure-neutral; it was unknowingly levered to slope. The bug was diagnosed during Week 7 stress testing: a $+50$ bp PC2 (slope) shock produced a large positive P&L residual that should have been near zero for a properly factor-neutral portfolio. Once identified, the fix was to regenerate the full 1971–2024 position history using the current, post-refactor PCA loadings consistently throughout. Net Sharpe fell from 0.51 to 0.11 upon correction—the 0.40 difference is the slope-bet alpha that was masquerading as RV alpha.

7.2 Remaining Limitations

Transaction costs. The headline assumption of 0.02 bp per leg per event is aggressive. Total headline costs are $\$807\text{K} - \$314\text{K} = \$493\text{K}$ over 40 years. At a realistic 1 bp round-trip per leg for off-the-run Treasuries (0.5 bp per event), costs scale by $0.5/0.02 = 25\times$ to $\approx \$12.3\text{M}$ —roughly $15\times$ the total gross P&L of $\$807\text{K}$. The strategy is only viable at these transaction costs in an electronic on-the-run execution environment, or with signals of order $15\times$ larger.

Signal universe. Only the 3Y and 7Y maturity buckets are traded. Extending to the full 1Y–30Y grid would increase diversification, lengthen average holding periods (reducing turnover), and improve the signal-to-cost ratio.

Stale hedge weights during open positions. Hedge weights are solved at position entry and held fixed until exit. As rates move over the holding period, the PCA exposures of the signal bond drift while the hedge stays constant, generating residual factor exposure that accumulates

gradually. This is distinct from the stale-loadings bug above (which used the wrong eigenvectors entirely); here the eigenvectors are correct but the hedge notionals are not re-optimized as the bond’s duration changes. Periodic intra-position re-solving (e.g., weekly) is the correct fix and would likely reduce the rising-rate and steep-curve Sharpe drag shown in Table 2.

Par-zero basis. The z-score residual is $y^{\text{par}} - y^{\text{zero}}(\tau)$, mixing true richness/cheapness with the par-zero basis. A more precise signal would use stripped zero rates throughout.

8 Conclusion

We have built a complete, reproducible Treasury relative-value pipeline: data ingest, Svensson curve fitting, PCA decomposition, signal generation, factor-neutral portfolio construction, and realistic backtesting with carry, repo, and transaction costs—all from a single YAML config. Three conclusions:

1. The mean-reversion signal is real. Gross Sharpe of 0.29 over 40 years is statistically credible and the hit rate is essentially coin-flip (50%), consistent with a low-information-content but directionally correct predictor.
2. The PCA hedge works as designed in most regimes. High-volatility and historically “risky” environments (2013, 2020, 2022) are not the primary failure modes; stale hedge weights in trending markets are.
3. Transaction costs are the governing constraint. At any realistic off-the-run execution cost the strategy loses money. The economically viable version of this strategy requires on-the-run execution, a much wider signal universe, or both.

References

- [1] Gürkaynak, R. S., Sack, B., and Wright, J. H. (2007). The U.S. Treasury yield curve: 1961 to the present. *Journal of Monetary Economics*, 54(8), 2291–2304.
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